



Society of Petroleum Engineers

SPE-230005-MS

Transforming Concepts into Reality: Integrating Pilot Vertical Wells with 3D Grid Modeling to Forecast Multistage Fracturing Outcomes in Gas Field

A. B. Pramadana, SLB, Pakistan; N. Hamim, Oil and Gas Development Company Limited, Pakistan; S. Kurniadi, S. Adiibah, A. S. Zaki, A. Ahmed, C. Tan, and H. Siddiqui, SLB, Pakistan

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230005-MS](https://doi.org/10.2118/230005-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The development of tight carbonate reservoirs presents a unique challenge due to their intrinsic heterogeneity, laminated facies, and stress anisotropy, which often complicate fracture growth, containment, and conductivity. In such environments, reliable stimulation through multistage hydraulic fracturing (MSF) requires robust subsurface characterization and accurate geomechanical modeling. However, constructing comprehensive three-dimensional (3D) Mechanical Earth Models (MEM) is rarely feasible in data-limited basins such as those in South Asia, where constraints on regional core coverage, stress testing, and seismic inversion limit subsurface understanding. As a result, operators face high uncertainty in fracture modeling and production forecasting.

This study introduces a novel, cost-effective workflow designed to overcome these challenges by integrating a core-calibrated one-dimensional (1D) MEM with an existing 3D static geological grid, anchored by fracture diagnostics and production data from a hydraulically fractured pilot vertical well. The workflow provides a pseudo-3D MEM that captures key stress and rock property variations within the reservoir, enabling accurate prediction of fracture geometry and reservoir performance despite limited regional datasets.

The pilot well served as the cornerstone of this workflow, offering essential insights into stress regimes, fracture containment, and inflow contribution from distinct Geological Flow Units (GFUs). Core calibration reconciled log-derived elastic properties with laboratory measurements, while image log analysis refined rock failure envelopes. Diagnostic fracture injection testing (DFIT) provided critical anchors for minimum horizontal stress. Post-fracture surveillance, including temperature logs, confirmed fracture height growth across the most productive GFU-3 intervals and validated the reliability of the MEM. These results were upscaled into a 3D static grid, creating a pseudo-3D framework capable of guiding MSF well design.

Building on pilot learnings and analogue benchmarking from the Field Q in southern Pakistan, a slanted horizontal MSF well was strategically planned parallel to the minimum horizontal stress direction to encourage transverse fracture development. The hydraulic fracturing design incorporated channel fracturing and dual proppant deployment—spherical intermediate-strength proppant (ISP) combined with rod-shaped proppant in tail-in stages—to enhance conductivity, mitigate proppant flowback, and accelerate post-

frac cleanup. Advanced reservoir simulations embedded these fracture properties within a sector model, forecasting multiple production scenarios based on varying bottomhole flowing pressures (BHFP).

This study contributes a practical, transferable methodology for operators working in data-limited carbonate reservoirs. By maximizing the value of pilot wells, propagating 1D MEMs into pseudo-3D grids, and embedding fracture diagnostics into simulation models, the approach reduces subsurface uncertainty and enables optimized MSF design. The findings offer critical insights for field development strategies in Pakistan and beyond, providing a blueprint for unlocking tight carbonate reserves where full-scale 3D MEM construction is impractical.

Introduction

Carbonate reservoirs are among the most prolific hydrocarbon-bearing systems worldwide, yet their development continues to pose significant challenges. The intrinsic heterogeneity of carbonate rocks, coupled with laminated facies and stress anisotropy, often complicates fracture propagation, containment, and conductivity. These challenges are particularly pronounced in tight carbonate gas reservoirs, where production enhancement is highly dependent on effective stimulation. Multistage hydraulic fracturing (MSF) has therefore emerged as a critical technology to unlock economic production. However, its success is strongly influenced by the quality of subsurface characterization, particularly the development of reliable geomechanical models capable of predicting fracture geometry and deliverability.

Accurate hydraulic fracturing design requires a Mechanical Earth Model (MEM) that integrates core, log, and field measurements to describe in-situ stress regimes and rock mechanical properties. Ideally, such models are built in three dimensions to capture both vertical and lateral heterogeneity. In practice, however, data constraints often limit model development, particularly in South Asian reservoirs where comprehensive 3D datasets are rarely available. The lack of such information increases uncertainty in fracture modeling, complicates production forecasting, and hinders the design of efficient MSF operations.

This study addresses these limitations by developing a cost-effective workflow that integrates core-calibrated one-dimensional (1D) MEM data with an existing three-dimensional (3D) static geological grid. The workflow leverages fracture diagnostics and production history from a hydraulically fractured pilot vertical well to calibrate and validate the integrated model. The methodology is applied to one of Pakistan's largest carbonate gas reservoirs, aiming to optimize horizontal MSF design and provide a transferable framework for other data-limited environments.

Geological and Geographical Background

The study area lies in the Western Fold Belt of Pakistan, within the Middle Indus Basin, a region that hosts several prolific carbonate gas reservoirs. The SXX Main Limestone (SXX-ML) is the primary producing formation and represents one of the most important gas-bearing intervals in the country. The reservoir was first explored in the mid-20th century, with commercial evaluation and field development accelerating in subsequent decades. The field is structurally complex, being compartmentalized into three major lobes—western, central, and eastern—each exhibiting variations in gas composition and heating value. Gas quality improves eastward, with the western lobe typically showing higher inert gas content and lower BTU value, while the eastern lobe delivers gas with the highest BTU rating (Ali et al., 2023). This regional variability not only influences gas marketing strategies but also complicates development planning, as stimulation outcomes can differ significantly between compartments. The location of the study area are illustrated in Figure 1.

Stratigraphically, the SXX-ML belongs to the Eocene sequence and is approximately 364 m thick. It conformably overlies the Upper Paleocene Dunghan Limestone and is sealed by the Lower Eocene Ghazij Shale, which provides an effective top seal for hydrocarbon entrapment. Internally, the reservoir is subdivided into four stratigraphic units (A–D). Most of the producible gas occurs in the upper three units (A,

B, and C), which collectively average ~175 m in thickness, whereas the lower unit (D) contributes minimally to flow (Ali et al., 2023). This vertical heterogeneity highlights the importance of selective perforation and targeted stimulation, as indiscriminate completion across the entire interval may lead to poor production efficiency.

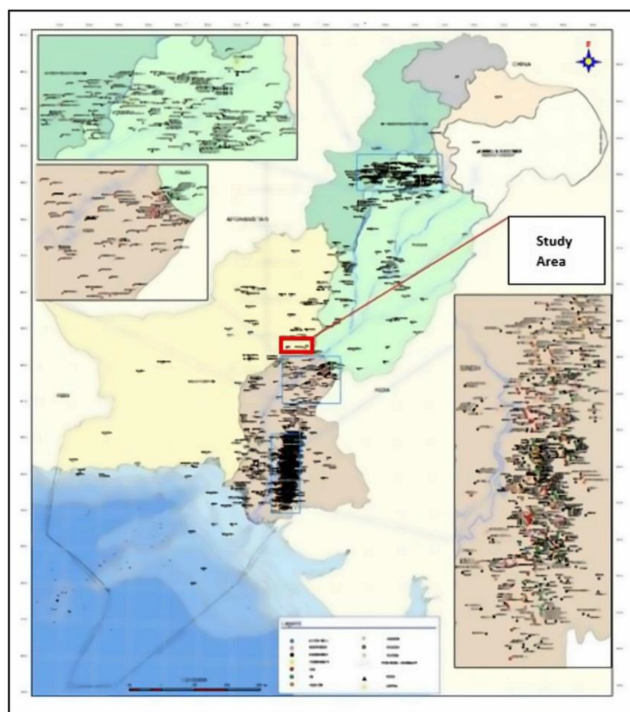


Figure 1—Location map of the study (after Ali et al., 2023).

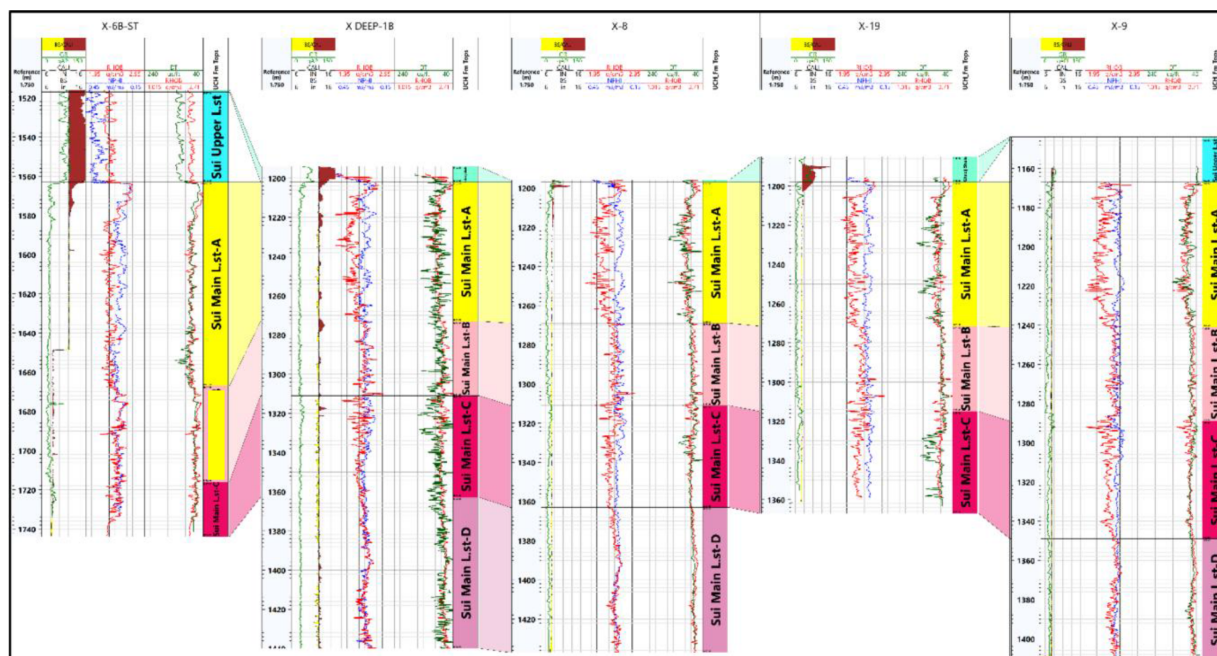


Figure 2—Stratigraphic column showing the SXX-ML reservoir, underlying Dunghan Limestone, and overlying Ghazij Shale seal, with internal units A–D (after Ali et al., 2023).

Petrophysical evaluation of the SXX-ML reveals significant variability in reservoir quality. More than 30 wells with core and wireline log datasets have been studied, enabling calibration of porosity and permeability

distributions. The matrix porosity ranges from nearly tight (0%) to as high as 35%, while permeability spans from 0.01 mD to 40 mD, depending on facies and degree of cementation. Despite intervals of relatively high porosity, many sections remain tight due to low effective permeability, indicating that porosity alone is an insufficient predictor of productivity (Ali et al., 2023). Core-log crossplots illustrate this variability, with data showing wide scatter between porosity and permeability, a signature of the heterogeneous depositional and diagenetic history of the formation.

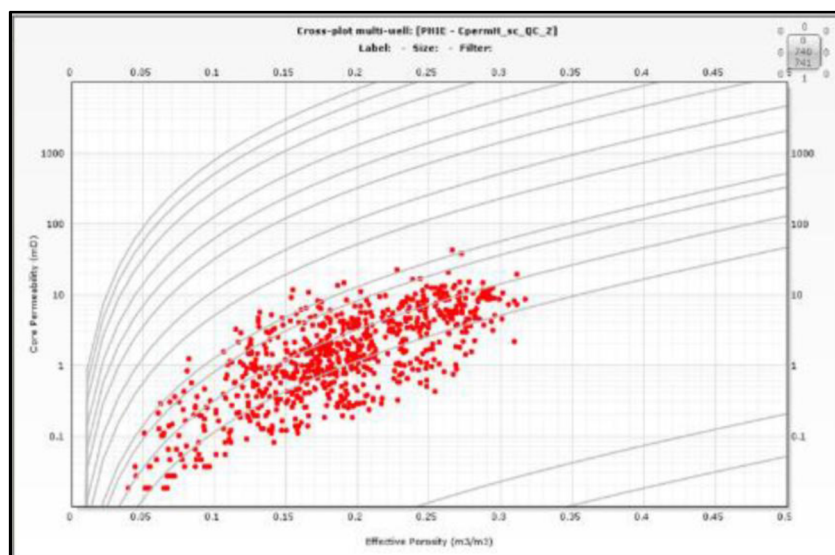


Figure 3—Core permeability versus porosity crossplot for SXX-ML, illustrating wide heterogeneity and variability in reservoir quality (after Ali et al., 2023).

To better classify reservoir performance, the concept of Geological Flow Units (GFUs) has been applied to the SXX-ML. Rock typing, based on porosity–permeability relationships, flow zone indicators (FZI), and production logging data, has enabled the subdivision of the formation into three GFUs with distinct reservoir qualities. GFU-1 comprises highly cemented limestone, characterized by low porosity, negligible permeability, and very poor reservoir quality. GFU-2 represents moderately cemented limestone with medium porosity and limited flow potential, while GFU-3 consists of poorly cemented limestone with higher porosity and permeability, offering the best reservoir quality, see comparison in Table 1 below. This classification has been validated by production logging, which consistently demonstrates that the majority of inflow originates from GFU-3 intervals. Conversely, GFU-1 and GFU-2 contribute only marginally, even when perforated (Ali et al., 2023). These findings underscore the importance of integrating geological facies and flow unit classification into hydraulic fracturing design, ensuring that stimulation energy is directed toward the most productive zones. The GFU classification and its validation through production logging are shown in Figure 4.

Table 1—Geological Flow Units (GFUs) Cut-offs (after Ali et al., 2023).

GFU	Porosity Range (%)	Permeability Range (mD)	Swir (%)	Geological Description
GFU-1	0 – 11	0.01 – 0.8	> 30	Highly cemented limestone
GFU-2	11 – 19	0.1 – 8	15 – 30	Moderately cemented limestone
GFU-3	19 – 35	1 – 40	< 15	Poorly cemented limestone

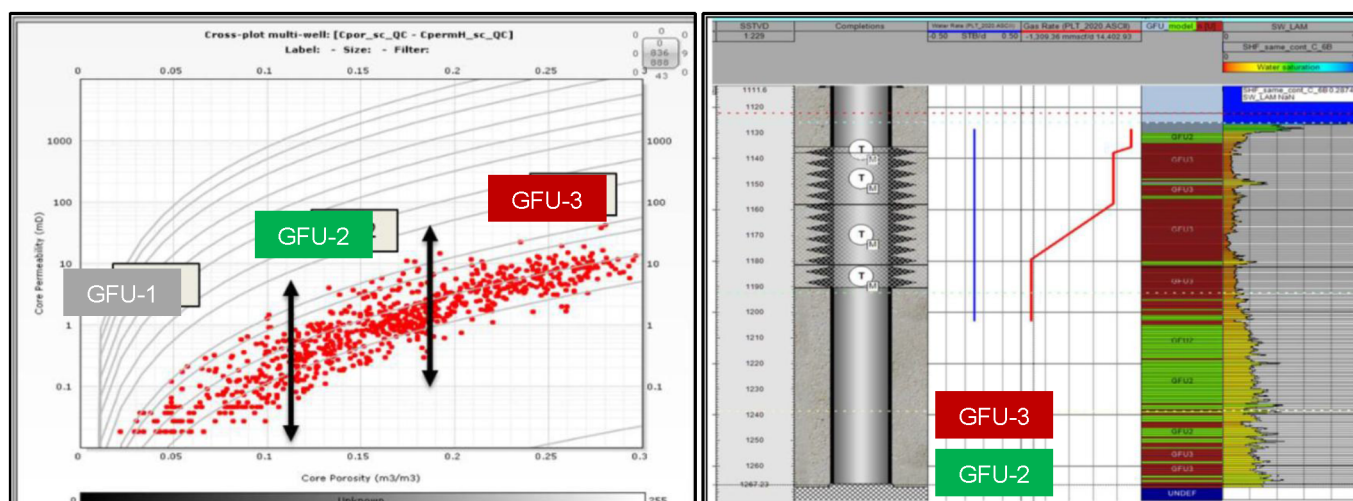


Figure 4—Geological Flow Units (GFU-1 to GFU-3) defined by porosity–permeability ranges and validated by production logging, showing inflow dominance from GFU-3 (after Ali et al., 2023).

In both the pilot and analogue wells, GFU-3 has consistently emerged as the primary target for stimulation, as it represents the most productive reservoir quality. Notably, in the analogue pilot well, GFU-3 was predominantly located in the upper section of the SXX-ML, where fracturing operations were successfully executed. This success provided an important milestone and a critical proof of concept before advancing toward full multistage fracturing (MSF) development. Equally important, the analogue case demonstrated that integrated data analysis—combining geomechanical modeling, core calibration, and fracture diagnostics—was essential to validate fracture placement and performance. These learnings highlight that robust pre-MSF workflows are indispensable in reducing uncertainty and ensuring that stimulation efforts in SXX-ML are focused on the most productive flow units.

Multistage Fracturing Approach Informed by the Vertical Pilot Well

The vertical pilot well served as a critical step in validating the geomechanical characterization of the SXX-ML reservoir and anchoring production forecasts. Post-DataFRAC surveillance confirmed that fracture height growth extended across the entire Upper section, where GFU-3 is predominantly developed, while remaining effectively contained within shale boundaries. As shown in Figure 5, the temperature log reveals a continuous cooled interval spanning all upper limestone, demonstrating multilayer connectivity and providing strong confidence in the calibrated stress model. These results confirmed that the core-calibrated 1D MEM, when upscaled into the 3D static grid, could reliably predict fracture geometry in this heterogeneous carbonate system.

A strong analogue to this outcome is the Pilot Well in southern Pakistan, where a comprehensive pilot program integrated DataFRAC, temperature logging, and channel fracturing to successfully overcome the transmissibility limitations of conventional proppant packs (Siddiqui et al., 2023). Operational diagnostics defined a fracture-gradient window of approximately 0.69 psi/ft. Calibration runs and temperature-log evaluation confirmed a fracture height of ~180 ft. The main treatment successfully placed ~130,000 lb of 30/50 and 20/40 proppant at ~21 bbl/min, reaching a peak concentration of ~6 PPA with a total fluid volume of ~2,000 bbl. This pilot demonstrated the effectiveness of channel fracturing in achieving reliable fracture geometry and conductive pathways in tight carbonate formations.

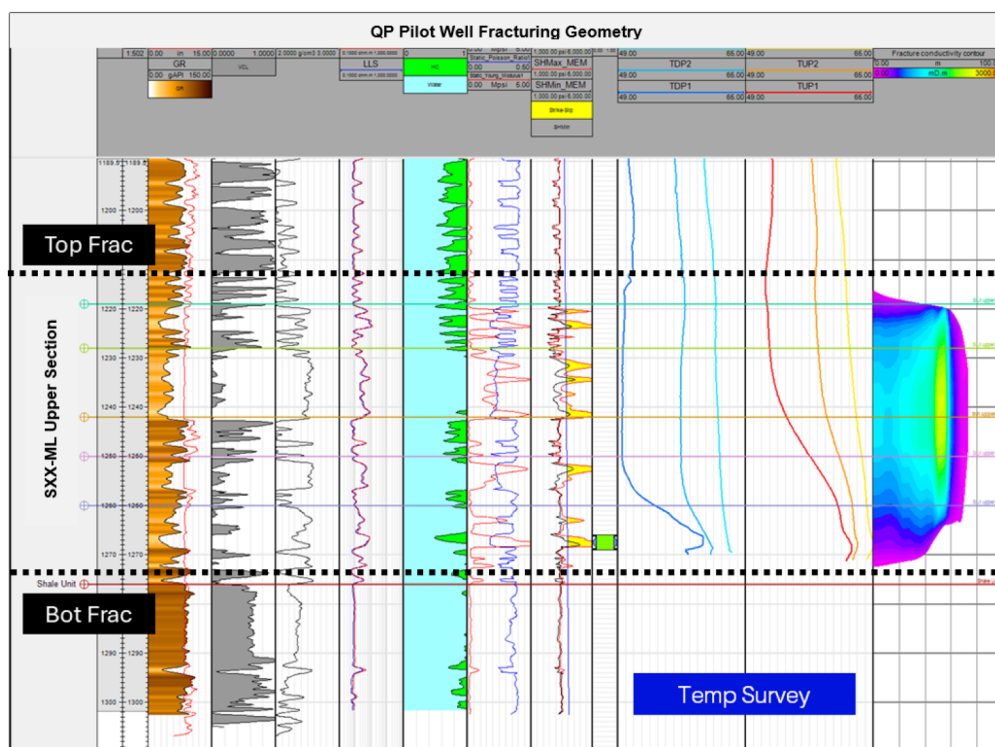


Figure 5—Pilot temperature-log height survey (post-DataFRAC). Fracture height growth covering SXX-ML upper section.

The production response was transformative. The Pilot shifted from uneconomic "bubble" flow (~ 0.06 n MMscf/d pre-frac) to sustained commercial output. The Pilot well initially delivered ~ 2.6 – 3.7 n MMscf/d, followed by a stable plateau of ~ 2.7 n MMscf/d throughout its first year, before gradually declining toward ~ 2.0 n MMscf/d by late 2024, as shown in Figure 6. This performance highlights the durability of fracture conductivity and the enhanced cleanup efficiency achieved through fiber-assisted channel fracturing.

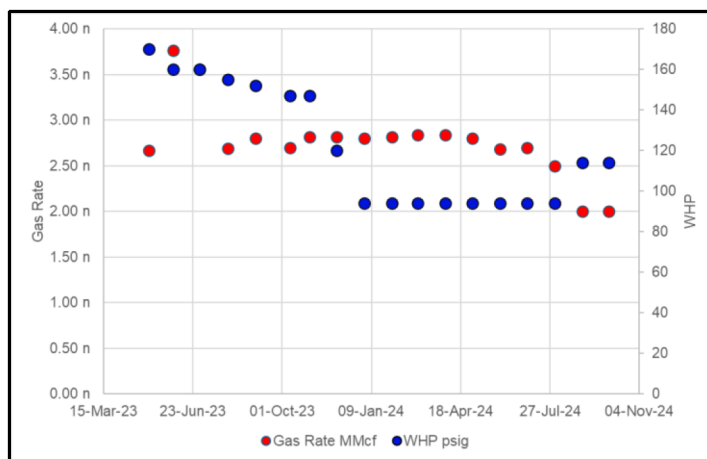


Figure 6—Post fracturing production performance of Pilot well.

These lessons directly inform the next stage of development. The demonstrated height growth across SXX-ML Upper in the pilot, coupled with the conductivity gains observed, point to a slanted MSF well as the optimal development strategy. The wellbore will be oriented parallel to the minimum horizontal stress to encourage transverse fractures, with stages distributed to cover all upper limestone. Stage placement will be optimized to maximize stimulation of GFU-3 intervals, which dominate inflow, while limiting unnecessary

treatment of poorer-quality GFU-1 and GFU-2 rock. Where transmissibility is constrained, fiber-assisted channel fracturing will be deployed to maintain near-wellbore conductivity and accelerate post-frac cleanup.

Together, the vertical pilot well highlight the value of an integrated workflow: pilot diagnostics + MEM calibration + analogue benchmarking. This approach reduces uncertainty, enables fit-for-purpose fracture designs, and establishes a robust technical foundation for full-field horizontal MSF deployment in Pakistan's carbonate reservoirs.

Multistage Fracturing Design Methodology

Rationale for a Simplified Geomechanical Framework

Hydraulic fracturing design in heterogeneous carbonate reservoirs depends fundamentally on an accurate geomechanical description. In an ideal scenario, this would be achieved through a fully developed 3D Mechanical Earth Model (MEM). However, in many mature or data-constrained basins, the cost, data requirements, and time associated with building a full 3D MEM are prohibitive. In this study, the challenge was further amplified by the lack of regional stress tests, limited offset core coverage, and sparse seismic-derived mechanical attributes.

To progress development without delaying the project for a multi-year geomechanical campaign, a pilot-anchored pseudo-3D workflow was adopted. The guiding principle was pragmatic: when the planned multistage fracturing (MSF) well lies within a few hundred meters of a hydraulically fractured pilot, and both wells share the same structural lobe and stratigraphic framework, the geomechanical description of the pilot can be reasonably extended laterally to the planned trajectory. This transfer assumes that stress orientation, pore pressure, overburden, and barrier behavior vary smoothly at the reservoir scale—an assumption supported by both the regional tectonic setting and surveillance data acquired in the pilot.

Drilling Plan to Optimize MSF Well

Efficient transverse fractures in horizontal wells are a critical factor in maximizing productivity in tight-gas developments. As shown in Figure 7 (Nicolaysen et al., 2015), the relative orientation of the wellbore to the in-situ stress field governs whether fractures propagate longitudinally along the wellbore or transversely across it. Transverse fractures, oriented orthogonally to the wellbore, are generally more efficient because they provide direct connection into the reservoir, creating a high-conductivity path with minimal near-wellbore pressure losses. Nicolaysen et al. (2015) noted that such fractures require a discrete single-point of entry for fluid during injection and production, which helps ensure consistent fracture growth and avoids complexity associated with uneven initiation from multiple perforation clusters.

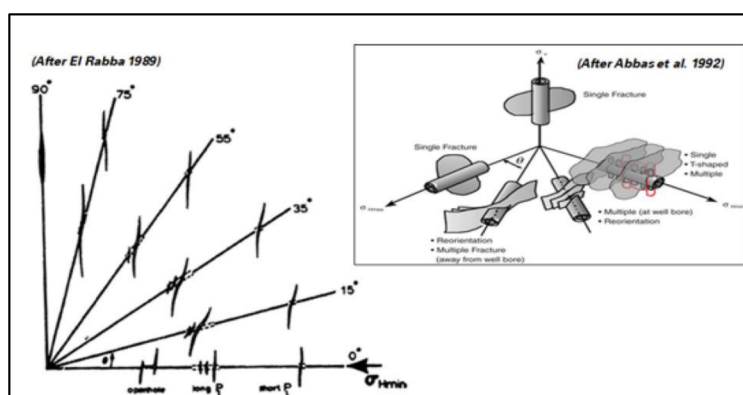


Figure 7—Fracture complexity and wellbore orientation relative to the in-situ stress field. Transverse fractures, when oriented orthogonally to the horizontal wellbore, promote a more efficient flow regime and enhanced reservoir connectivity (after Nicolaysen et al., 2015; adapted from El Rabaa, 1989 and Abbas et al., 1992).

Moreover, the study highlighted that radial-convergent flow from a single-point entry can cause excessive drawdown, drag force, and elevated fluid velocity near the wellbore if not managed properly. To reduce this negative impact on performance, industry consensus cited in the paper recommends limiting wellbore connection intervals to $\leq 4\times$ open-hole diameter. This balance between wellbore connection design and stress alignment is essential to sustain long-term productivity in horizontal wells. For the MSF planned well case, the maximum horizontal stress azimuth is approximately N165E, with SHmin direction around 75°. Both proposed trajectories at 85° and 87.5° azimuth are well within the optimal stress window for initiating transverse fractures. Such alignment ensures more predictable fracture initiation, efficient reservoir connection, and higher overall stimulation effectiveness (Nicolaysen et al., 2015)

Design Methodology

The MSF Planned Well, located only ~300–400 m from the pilot (Figure 8), lies entirely within this influence zone. In this context, the pilot's fracture diagnostics, stress calibration, and core-derived rock properties provide a robust and reliable anchor. By contrast, such an approach would not be appropriate for larger step-outs into different compartments, across faults, or into structurally distinct lobes, where stress rotation and pore-pressure variation would necessitate a full 3D MEM. In this case, however, the pilot-anchored pseudo-3D method provides the right balance of technical rigor and operational practicality.

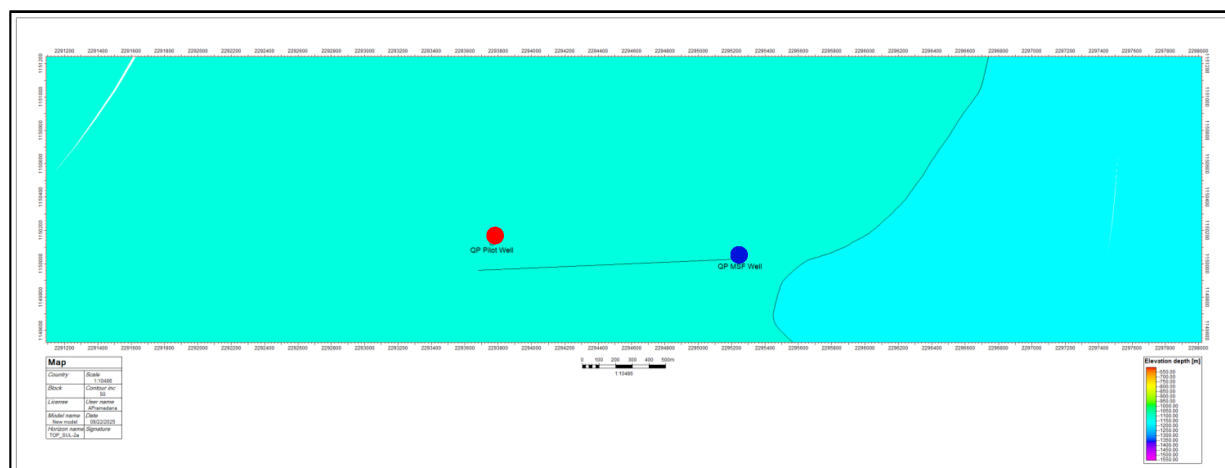


Figure 8—Pilot vertical well and MSF well location, ~300–400 m apart.

Accordingly, the methodology followed a structured three-part workflow:

1. Build a core-calibrated 1D MEM from the pilot well.
2. Upscale and distribute the 1D MEM properties into the static geological grid to form a pseudo-3D MEM.
3. Apply advanced fracture modeling (to simulate fracture geometry and conductivity) coupled with advanced reservoir modeling (to forecast well performance and optimize design).

Build a Core-calibrated 1D MEM from the Pilot Vertical Well. The vertical pilot well provided the foundation for developing a core-calibrated 1D Mechanical Earth Model (MEM) that could capture the SXX-ML reservoir's complex geomechanical behavior with a high degree of confidence. The first step involved reconciling wireline log data with core-derived elastic properties. As expected in heterogeneous carbonates, the uncorrected sonic-derived dynamic elastic properties significantly overestimated stiffness compared to static laboratory results. To address this, compressional and shear slowness were, scaled as per Table 2, shifting log-derived Young's modulus and Poisson's ratio into alignment with core tests. This correction was essential to avoid artificially high stiffness values, which would otherwise result in unrealistic

fracture toughness and underestimated rock deformability in modeling. The calibration is illustrated in Figure 9, which shows the comparison of uncorrected log-derived dynamic moduli with corrected static equivalents, highlighting the improvement achieved by core matching.

Table 2—Parameters used for 1D MEM in Pilot well.

Original Parameter	Adjustment Coefficient	Equation		Remark
DTC	0.8	-		-
DTSH	0.7	-		-
E_{sta}	-	$E_{sta} = 0.1229876 \times (E_{dyn})^{1.97502}$		Carbonate
E_{sta}	-	Lacy		Shale
ν_{sta}	-	$\nu_{sta} = 0.6 \times \nu_{dyn}$		Carbonate
ν_{sta}	-	$\nu_{sta} = 0.9 \times \nu_{dyn}$		Shale
UCS	-	Chang		Carbonate
UCS	-	Hosrud		Shale
TSTR	-	$TSTR = 0.12 \times UCS$		-

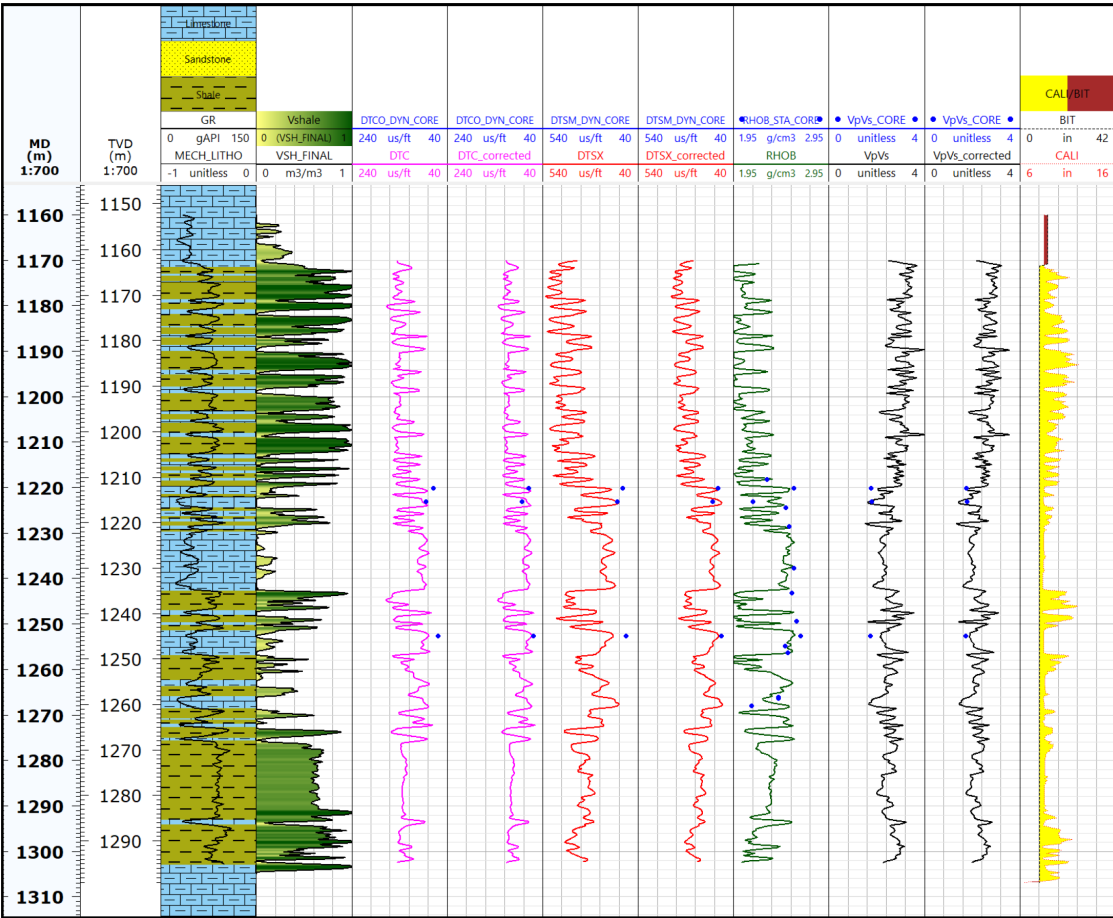


Figure 9—Comparison of dynamic vs. static elastic properties in the SXX-ML. Sonic log-derived Young's modulus and Poisson's ratio adjustment with the core test.

Once the elastic calibration was completed, static elastic properties were derived using established empirical transforms. For the carbonate reservoir units, a power-law correlation (Table 2) provided the closest match with core measurements, capturing the nonlinear stress–strain behavior typical of limestone

formations (Collins, 1987). In shales, where calibration data were limited, the Lacy (1997) transform was applied to convert dynamic to static modulus. Poisson's ratio was also corrected based on core test evidence, as summarized in Table 2, ensuring that the MEM properly reflected the rock's elastic deformability. Collectively, these transforms showed strong agreement with measured core data and established a consistent elastic framework for the MEM, as illustrated in Figure 9.

Building on this framework, rock strength and failure parameters were incorporated. For carbonates, UCS was estimated using the Chang et al. (2006) correlation, while shale UCS was calculated from the Horsrud (2001) relation. Tensile strength was assumed to be 12% of UCS, consistent with published laboratory data, and the initial friction angle was derived from gamma-ray–based lithology correlations. To improve accuracy, these values were refined using borehole image logs, ensuring that the MEM's failure envelopes aligned with observed breakout zones.

The calibrated failure envelope, illustrated in Figure 10, highlights the predicted UCS and tensile strength trends across lithologies, together with Mohr–Coulomb envelopes validated against borehole breakout observations. Building on this foundation, stress and pore pressure profiles were constructed to complete the geomechanical description. Vertical stress was derived by density integration, yielding an average overburden gradient of ~0.95 psi/ft. Pore pressure was estimated using Eaton's method, constrained by mud weights between 1.1 and 1.34 s.g., and corroborated by reservoir pressure measurements that indicated ~39% depletion in the carbonate pay (see Table 2). The minimum horizontal stress was anchored using DFIT and DataFRAC closure pressures, while the maximum horizontal stress was constrained by drilling-induced fractures and breakout orientations. These diagnostics consistently indicated a maximum stress azimuth of approximately N165E, as summarized in Figure 11. The integrated stress and pore pressure tracks are presented in Figure 8, showing continuous depth profiles of vertical stress, minimum horizontal stress, maximum horizontal stress, and pore pressure that collectively validate the reliability of the calibrated MEM.

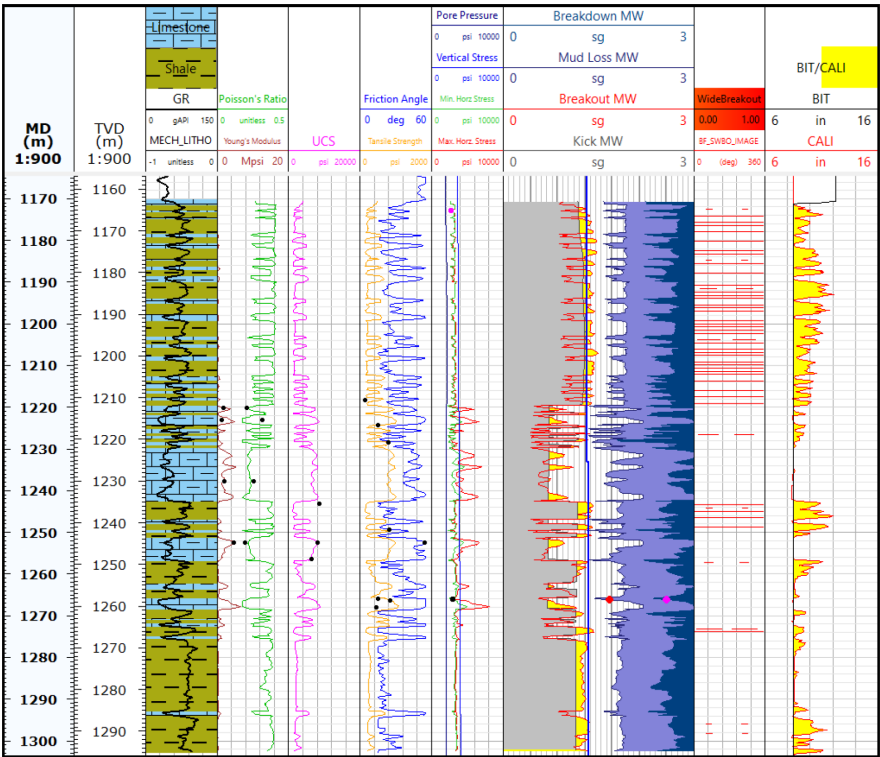


Figure 10—Calibrated 1D MEM for the SXX-ML. Elastic property transforms, UCS correlations, and stress–pore pressure profiles benchmarked to core data, breakouts, and DFIT/DataFRAC diagnostics.

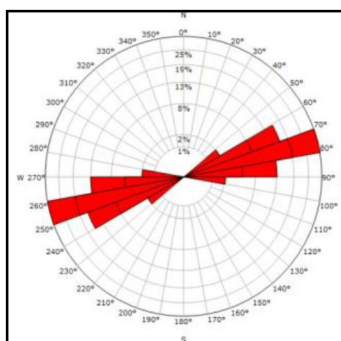


Figure 11—Breakouts trend N165E, indicating minimum horizontal stress is NW–SE and maximum horizontal stress is NE–SW.

Pseudo-3D MEM Construction from 1D Pilot Data. In an ideal scenario, multistage fracturing design is conducted using a fully coupled 3D framework that integrates a static reservoir properties grid with a dedicated 3D Mechanical Earth Model (MEM). This comprehensive approach captures spatial variability in in-situ stresses, pore pressure, and elastic properties, enabling high-confidence predictions of fracture geometry and production performance. However, constructing a field-scale 3D MEM requires extensive datasets—including multi-well core coverage, regional stress testing, seismic inversion, and cross-well calibration—which were not available for this study.

Instead, a practical alternative was adopted by leveraging the existing static reservoir grid (Figure 12), which includes key production modeling parameters such as porosity and permeability. A simplified workflow was implemented wherein the core-calibrated 1D MEM from the vertical pilot well was propagated into the static geological grid to form a pseudo-3D MEM. This decision was supported by the short offset (~300–400 m) between the pilot and MSF wells, both located within the same structural lobe and stratigraphic framework. At this scale, lateral variations in stress, pore pressure, and elastic properties are expected to be minor, making the pilot well a reliable calibration anchor.

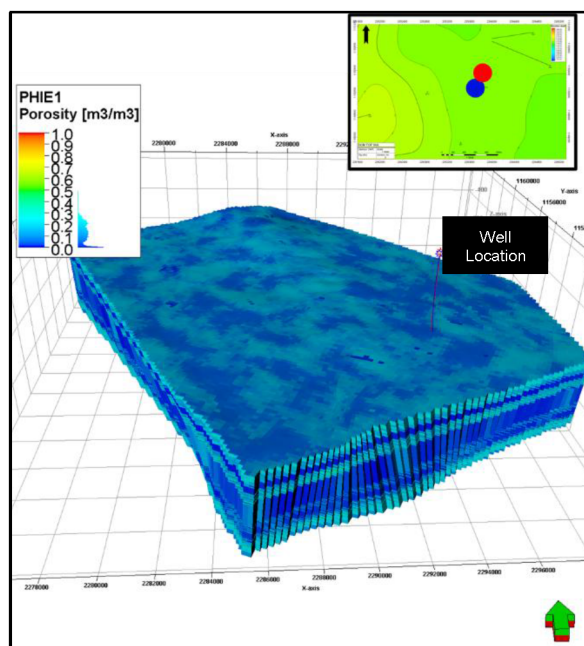


Figure 12—Overview of Static Grid with Well Location

Figure 13 contrasts the ideal integrated workflow (3D reservoir + 3D MEM) with the simplified method used in this study (3D reservoir + propagated 1D MEM). While the latter introduces some uncertainty in

lateral distribution, it remains technically defensible and operationally efficient for near-pilot development. The pseudo-3D MEM was constructed by stratigraphically distributing geomechanical properties from the pilot well across the static grid, preserving vertical heterogeneity and ensuring lateral continuity. The following parameters were propagated:

- **Minimum horizontal stress:** Anchored using DFIT and DataFRAC closure pressures
- **Maximum horizontal stress:** Constrained by image log breakouts and drilling-induced fractures; orientation $\approx$ N165E
- **Vertical stress:** Derived from integrated density logs
- **Poisson's ratio:** Converted to static values using core-calibrated transforms
- **Young's modulus:** Calibrated to static conditions from core measurements
- **Pore pressure:** Estimated via Eaton's method and bounded by observed mud weights

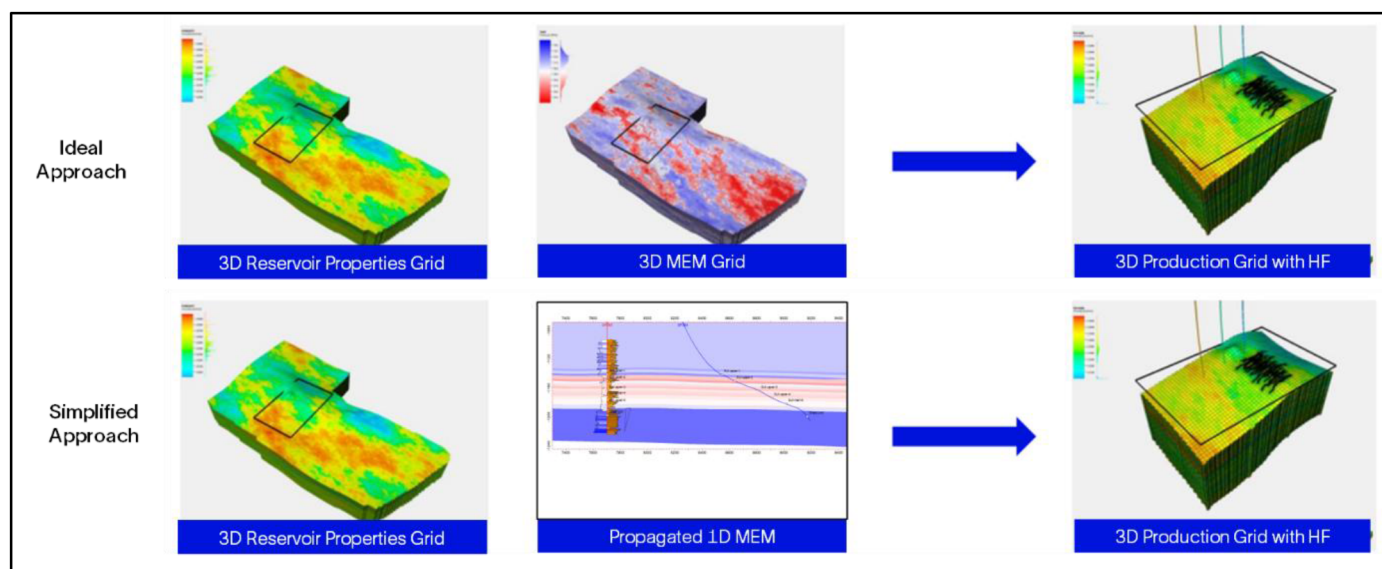


Figure 13—Illustration of the ideal integrated workflow (3D reservoir + 3D MEM) versus the simplified method used in this study (3D reservoir + propagated 1D MEM).

Property propagation was performed along stratigraphic layers corresponding to reservoir units, generating a laterally continuous pseudo-3D MEM that serves as the foundation for fracture simulation and treatment design. Figure 14 presents a representative cross-section showing the spatial distribution of minimum horizontal stress; similar propagation was applied to all other geomechanical parameters. While suitable for near-offset development, this method would not be appropriate for larger step-outs into different compartments, across faults, or into structurally distinct lobes, where stress rotation and pore-pressure variation necessitate a full 3D MEM. In this case, however, the pilot-anchored pseudo-3D approach strikes an effective balance between technical rigor and operational practicality. Once the MSF well was drilled, the model was updated using actual log data from the lateral, further refining geomechanical property distributions and improving the accuracy of stage placement and fracture modeling.

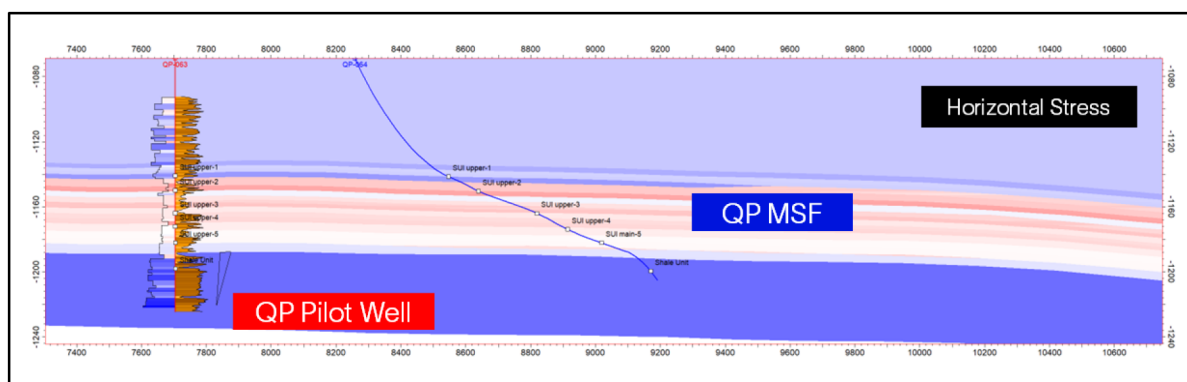


Figure 14—Cross-sectional view of the pseudo-3D Mechanical Earth Model (MEM) showing spatial distribution of minimum horizontal stress across the reservoir.

Fracture and Reservoir Modeling for Design Optimization.

Stage and Cluster Placement Strategy

The MSF well was designed using a pseudo-3D Mechanical Earth Model (MEM) integrated with reservoir quality and completion feasibility indicators. The well trajectory was planned as a slanted lateral to maximize contact with the Upper SXX-ML interval while minimizing drilling complexity. Based on pilot surveillance and analog performance, five fracture stages were proposed—one per sub-stratigraphic limestone unit—to ensure full interval stimulation and treatment efficiency. Stage selection was guided by a Reservoir Quality Index (RQI), incorporating thresholds for resistivity ($>1.0 \text{ ohm}\cdot\text{m}$), water saturation (Completion Quality Index (CQI) was also applied to exclude intervals with fracture gradients below 0.8 psi/ft and to classify the prevailing stress regime. This approach also include Completion Class Quality Index (CCQI) that favors normal faulting stress environments, which are more conducive to vertical fracture propagation and proppant placement, thereby enhancing stimulation effectiveness.

The design methodology followed the Completion Advisor workflow (Cipolla et al., 2011), integrating petrophysical and MEM-calibrated data to optimize fracture placement based on stress contrast, brittleness, and porosity. Logs from the Pilot well were upscaled into the 3D grid to define Stimulation Quality Index (SQI) layers, combining porosity, resistivity, Young's modulus, Poisson's ratio, and minimum horizontal stress. Stage spacing was dynamically adjusted using adaptive cluster spacing principles (Walker et al., 2012), with wider spacing (60–75 m) in low-transmissibility GFU-1 zones and tighter spacing (30–50 m) in productive GFU-3 layers. This reservoir-centric strategy minimized stress interference and avoided stimulation of non-contributing rock.

Figure 15 illustrates the preliminary stage selection, incorporating both reservoir quality and completion feasibility indicators, including stress regime classification. Field applications of RQI-CQI-based designs have shown that over 80% of perforation clusters contribute to flow in tight reservoirs (Babatunde et al., 2013; Ejofodomi et al., 2014). It is important to note that this design is preliminary and based on propagated pilot logs. Final stage and cluster placement will be refined using actual log data from the MSF well. Additional parameters—such as total gas content, permeability, porosity, and other reservoir-specific indicators—will be incorporated to further enhance stage selection and optimize stimulation outcomes.

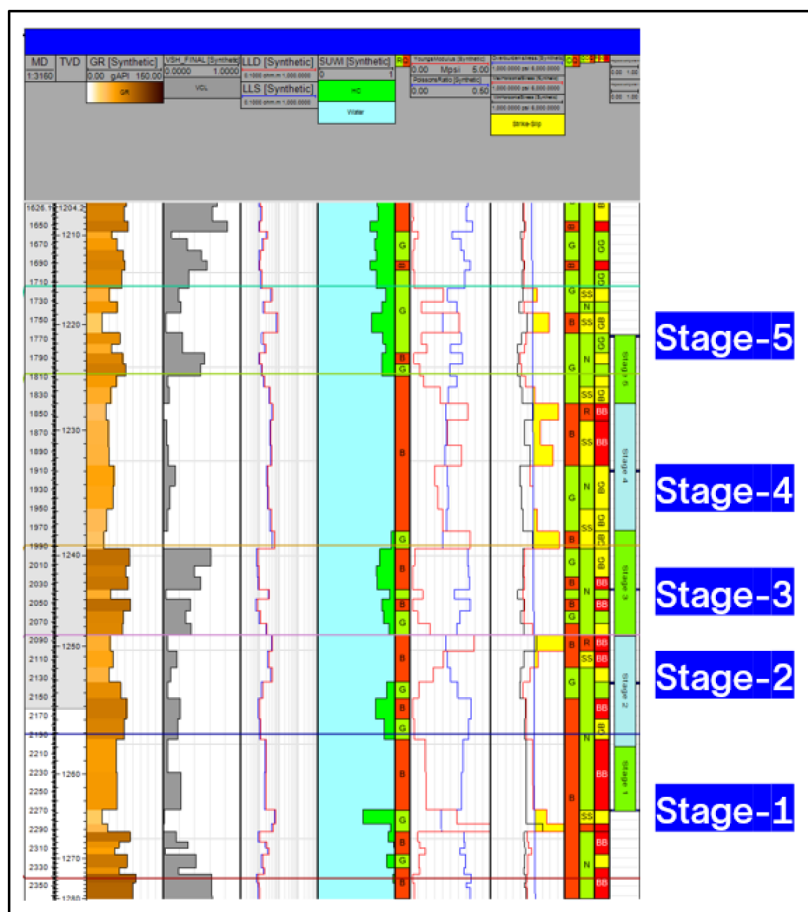


Figure 15—Preliminary fracture stage selection for the MSF well, integrating Reservoir Quality Index (RQI), Completion Quality Index (CQI), and stress regime classification.

Hydraulic Fracture Design

The hydraulic fracturing design for the MSF well incorporates two synergistic technologies: channel fracturing and a flowback control proppant strategy. This approach was adopted following observations of proppant flowback during the execution of the Pilot vertical well. By integrating these methods, the design aims to enhance fracture conductivity, optimize hydrocarbon recovery, and reduce operational risks such as screen-outs and proppant migration.

Channel fracturing is applied to enhance fracture conductivity and facilitate more efficient post-treatment cleanup. Unlike conventional designs that uniformly pack the fracture with proppant, this technique alternates fiber-laden sweeps with proppant slugs to construct discrete pillars and open flow channels within the fracture. These unpropped channels remain conductive, promoting improved hydrocarbon transport and minimizing gel residue accumulation. The method has demonstrated several operational advantages, including reduced proppant requirements, lower injection pressures, and improved fracture containment—particularly in formations characterized by narrow fracture apertures and suboptimal rock quality (Al Maki et al., 2024).

Proppant selection in the MSF design employs a dual-deployment strategy to optimize both injectivity during placement and conductivity retention over the well's production life. The initial stages utilize 20/40 mesh Intermediate Strength Proppant (ISP), selected for its spherical geometry. In the tail-in stages, rod-shaped proppant is introduced to reinforce near-wellbore conductivity and mitigate proppant flowback. The rod-shaped geometry promotes mechanical interlocking within the fracture, resulting in higher pack porosity and enhanced structural stability under drawdown conditions. Laboratory studies have shown that

rod-shaped proppants retain up to 93% of their conductivity after cleanup, significantly outperforming conventional spherical proppants, which retain only 65% (McDaniel et al., 2010).

Fracture sizing and proppant loading for the MSF well are calibrated through a combination of reservoir simulation and field-derived performance data. Each stage is designed to stimulate approximately 40 meters of net pay, with a calibrated proppant intensity of 2.5 metric tons per meter—resulting in a total proppant mass of roughly 100 metric tons per stage. This loading strategy aligns with established practices in analogous Pilot vertical well.

The fluid system deployed in the MSF well utilizes a crosslinked gel specifically engineered to support fiber transport and proppant suspension under reservoir conditions. Drawing on the formulation and performance data based on Siddiqui et al., 2024 – execution of Pilot well , the selected fluid 20 ppt gel loading demonstrated high viscosity retention (>100 cP) across a broad shear rate range, ensuring effective proppant placement accross the upper limestone and channel formation. Thermal stability was confirmed at bottomhole temperatures between 150°F and 290°F, with consistent rheological behavior throughout the pumping cycle as seen in Figure 16.

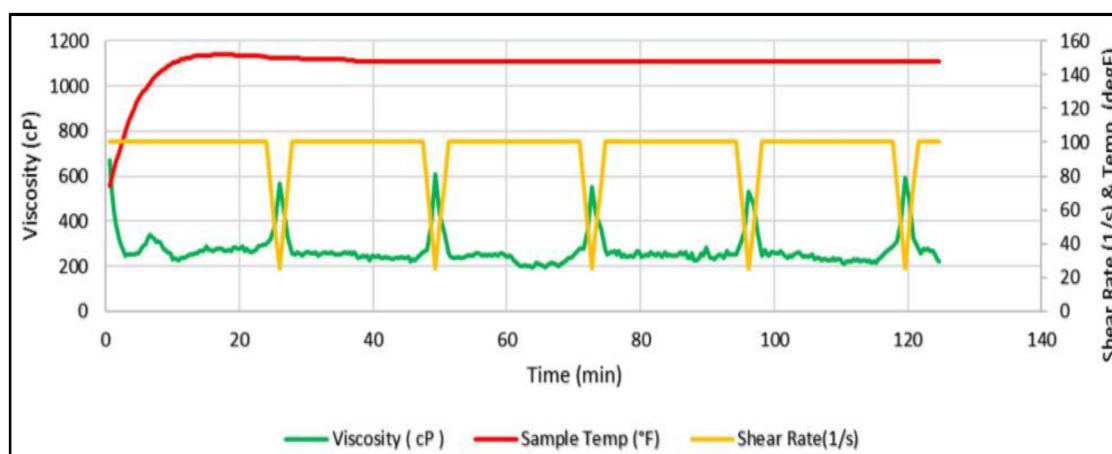


Figure 16—HPHT baseline fluid stability test (after Siddiqui et al., 2024)

A comprehensive summary of the MSF hydraulic fracturing design—including key stimulation parameters, fluid system rationale, and proppant deployment strategy—is provided in Table 3. The corresponding simulation outputs, which illustrate fracture geometry, proppant placement, and conductivity retention across stages, are presented in Figure 17 and Table 4. These visuals collectively validate the design's effectiveness in mitigating proppant flowback and optimizing stimulation performance across the targeted reservoir intervals.

Table 3—Fracturing design approach summary.

Design Element	Description	Purpose / Benefit	Reference
Fracturing Strategy	Channel fracturing with fiber	Enhances conductivity, reduces gel residue, lowers injection pressure	Al Maki et al., 2024
Proppant Strategy	Dual-deployment: 20/40 ISP (spherical) + rod-shaped proppant	Balances injectivity and conductivity; rod-shaped proppant resists flowback	McDaniel et al., 2010
Proppant Loading	~2.5 metric tons/meter over 40 m net pay (~100 metric tons/stage)	Ensures adequate fracture coverage without overdesign	Pilot vertical well data
Fluid System	Crosslinked gel, 20 ppt loading	Supports fiber/proppant transport; stable at 150–290°F; breaks cleanly post-treatment	Siddiqui et al., 2024
Observed Challenge	Proppant flowback in pilot vertical well	Informed adoption of flowback control strategy	Internal field observation

Table 4—Fracture geometry simulation output summary for MSF well. This table presents stage-wise simulated fracture dimensions, including net pressure, half-length, width, and height.

	Net Pressure (psi)	Prop Half Length (m)	Prop Width (mm)	Prop Height (m)
Stage-5	253	110.29	5.64	40.37
Stage-4	146	80.17	6.21	74.96
Stage-3	235	106.85	6.70	51.62
Stage-2	96	109.54	6.53	52.09
Stage-1	145	111.54	5.83	43.12

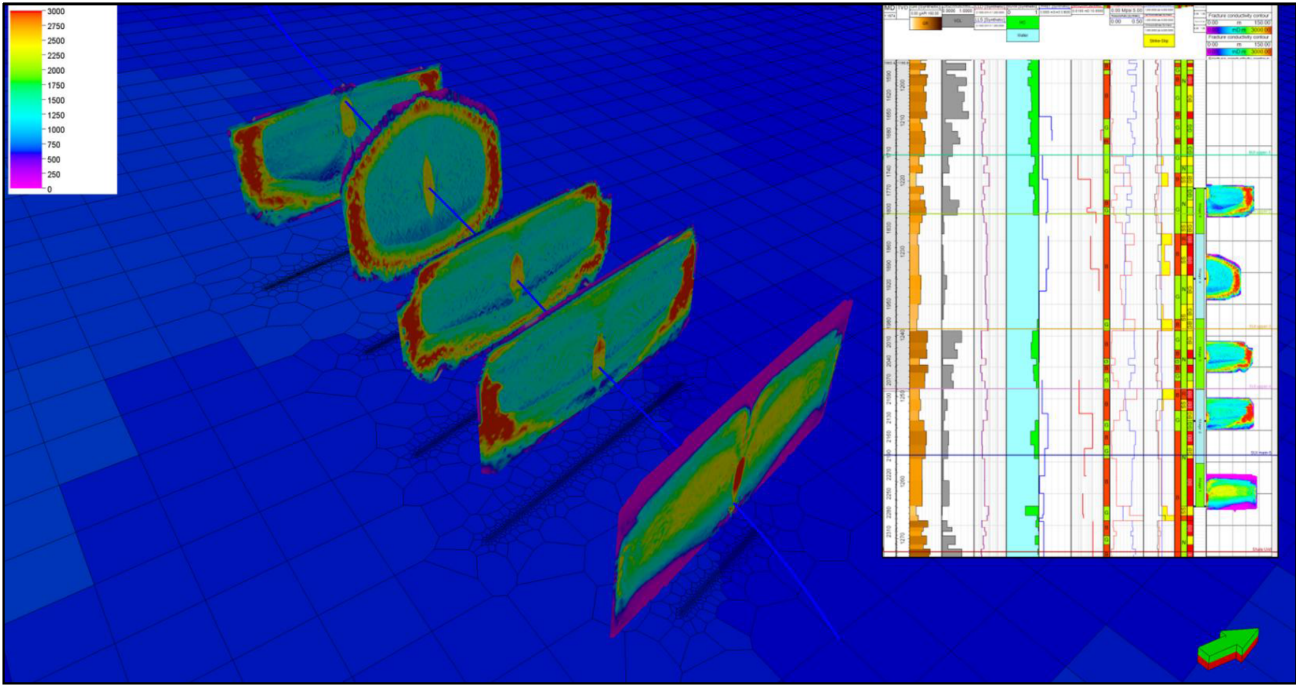


Figure 17—Simulated fracture geometry and proppant distribution across MSF stages.

Production Simulation Post Fracturing

A sector-based dynamic simulation model was developed to evaluate the production potential of well MSF under a five-stage hydraulic fracturing design. The approach integrated static model inputs, fracture

characterization, and bottomhole flowing pressure sensitivities, with the objective of forecasting post-fracture performance over a three-year horizon. The workflow drew on analogue well Pilot Well for calibration, ensuring consistency with observed production history and thereby reducing uncertainty in predictive outcomes. The fluid system was defined as dry gas, reflecting the absence of significant condensate or liquid dropout under expected reservoir conditions. Reservoir parameters such as porosity, permeability, and net-to-gross thickness were directly imported from the static geological model, which itself was calibrated against well logs and core data. These parameters formed the foundational grid properties that governed fluid flow within the simulation.

A key uncertainty in the forecast was the bottomhole flowing pressure (BHFP), which directly influences drawdown, productivity, and ultimate recovery. To capture this uncertainty, four scenarios were defined:

- Higher case : ~400 psi (optimistic, high drawdown support)
- High case : ~550 psi
- Mid case : ~700 psi (reference, aligned with Pilot Well)
- Low case : ~850 psi (conservative, limited drawdown)

Hydraulic fracture properties were explicitly embedded into the simulation grid (Figure 18), incorporating fracture geometries, propped half-lengths, heights, and conductivities derived from MSF fracture design outputs. These parameters were themselves constrained by geomechanical models and treatment feasibility assessments, ensuring realistic representation of fracture behavior. Accurate fracture modeling was essential for defining the stimulated reservoir volume (SRV) and capturing the influence of proppant-supported channels on well productivity.

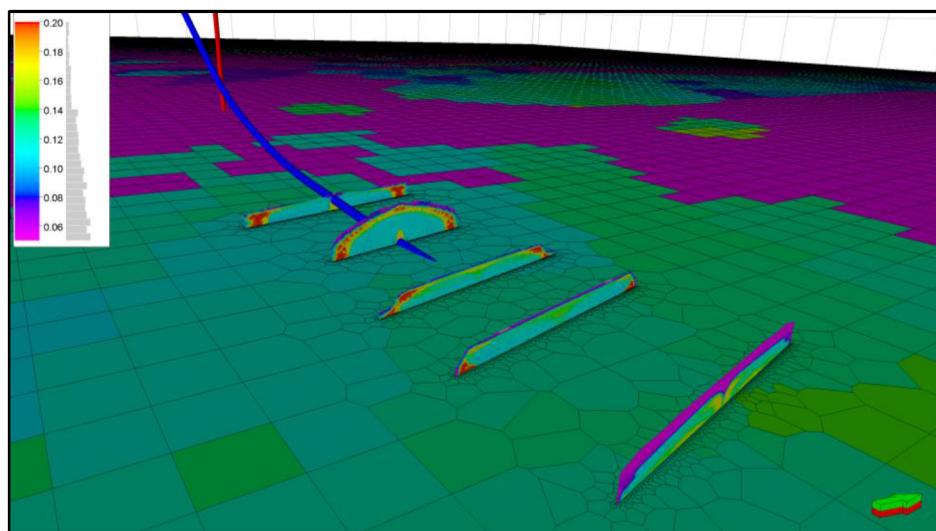


Figure 18—Embedded hydraulic fracture properties within the simulation grid.

To minimize computational bias and mitigate boundary-related artifacts, a sector model was employed, focusing on the 90-acre drainage area surrounding the MSF well. This corresponds to an effective radial extent of approximately 340 meters, allowing flow behavior to remain localized within the expected drainage region while preserving numerical efficiency. The simulation grid was constructed to distinguish between stimulated and unstimulated reservoir zones. As shown in Figure 19, active cells delineate the reservoir flow domain, whereas inactive cells serve as boundary buffers to prevent artificial pressure support. Grid resolution was strategically refined near the horizontal wellbore and fracture planes to accurately capture steep pressure gradients and transient flow behavior. In contrast, coarser grid cells were

applied in distal regions to reduce computational overhead without compromising model fidelity. This multi-scale gridding approach ensured that both near-wellbore dynamics and far-field reservoir responses were adequately resolved within the simulation framework.

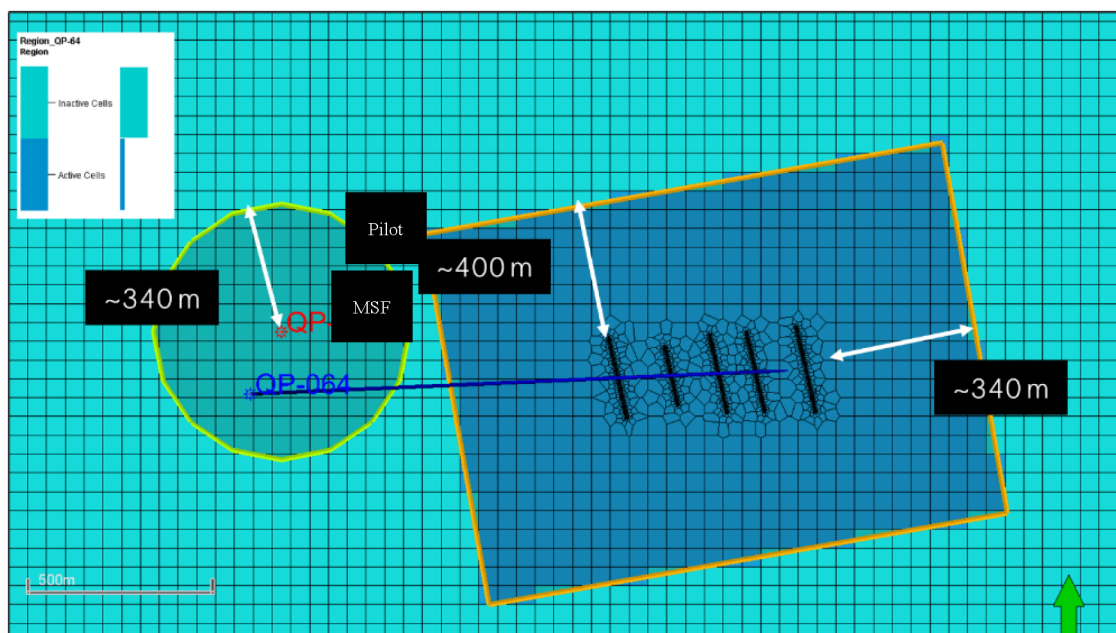


Figure 19—Sector model grid configuration for MSF simulation.

The MSF simulation model incorporated five discrete hydraulic fracture stages distributed along the horizontal wellbore. Each stage extended laterally by approximately 400 meters and was vertically constrained by natural stratigraphic barriers. These fractures were embedded into the reservoir grid using transmissibility multipliers, effectively simulating fracture-enhanced conductivity. As a result, the stimulated envelope provided connectivity to a significantly larger volume of gas-bearing rock than would be accessible in the unfractured state. To validate the model configuration, the setup was benchmarked against the Pilot sector model. The drainage radius for Pilot well was similarly estimated at approximately 340 meters, and its historical production behavior confirmed the reasonableness of this assumption for the reservoir under study.

With fracture geometries embedded and input sensitivities defined, the MSF well was forecasted over a three-year post-fracture production period. The simulation accounted for convergence effects typical of horizontally fractured wells, where localized high-velocity flow near fracture–wellbore intersections leads to accelerated early-time decline but supports sustained long-term recovery.

Immediately following fracture cleanup (time zero), the model predicted initial gas production rates (refer to Figure 20). As expected for tight gas systems, rates declined sharply during the first year due to high drawdown and limited pressure support. By the end of Year 1, production rates had decreased to a range of 0.82a–1.00a MSCF/d, and further declined to 50.27a–0.34a MSCF/d by Year 3. The spread between high and low cases underscores the sensitivity of production performance to flowing pressure assumptions. Despite the steep decline trend, cumulative production forecasts indicated substantial recovery potential. After one year, cumulative gas volumes ranged from 0.36b–0.46b BCF and reaching 0.79b–1.00b BCF by the end of Year 3. These results demonstrate the significant incremental contribution of the five-stage hydraulic fracturing program.

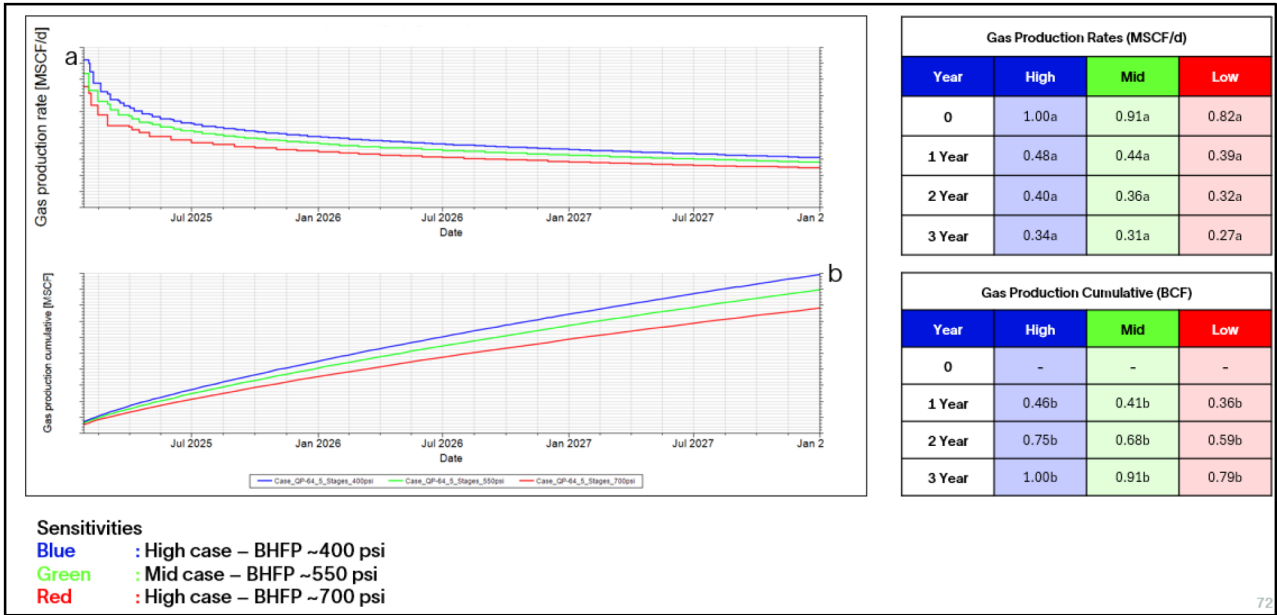


Figure 20—Forecasted Gas Production Rates and Cumulative Recovery for MSF Over a Three-Year Post-Fracture Period.

Key Insights and Implications. The modelling study highlights several critical insights:

- **Fracture Optimization** – The five-stage fracture design in MSF effectively connects a substantial drainage volume. This underscores the necessity of stage count and spacing optimization in tight-gas developments.
- **Pressure Sensitivity** – Bottomhole flowing pressure is a decisive parameter. Lower flowing pressures extend drawdown and accelerate recovery, but also intensify early decline. Effective reservoir pressure management strategies are required to balance these effects.
- **Analogue Validation** – The strong alignment of Pilot Well simulations with observed data provided confidence in applying the same methodology to MSF well, reducing uncertainty in forecasts.
- **Convergence Effects** – Horizontally fractured wells exhibit pronounced convergence behavior, characterized by localized high-velocity flow near fracture–wellbore intersections. This accelerates early-time decline rates while supporting sustained cumulative recovery. However, the magnitude of convergence can vary depending on fracture geometry and reservoir properties, and if not accurately captured, may lead to overestimation of production potential in simulation models.

This modeling approach is still considered preliminary, as it relies on a simplified 3D static grid that does not fully capture the complexity of the reservoir. Key features such as geological heterogeneity, fault systems, and dynamic fracture behavior were approximated rather than explicitly modeled. While the simulation includes important effects like pressure sensitivity and convergence, the static grid has not been calibrated against full-field production data. Ideally, the model should be history-matched to actual performance from multiple wells to validate its accuracy and improve confidence in the forecasts. Without this calibration, there is a risk of overestimating production potential, especially in areas where fracture behavior or reservoir quality varies significantly. Further refinement using detailed field data and geomechanical coupling would help reduce uncertainty and improve the reliability of long-term predictions.

Key Technical Gap

To support future development planning in the SXX-ML formation, a structured review was conducted to benchmark the current status of data acquisition, reservoir modeling, and stimulation design readiness

across critical domains. Foundational work—including 1D geomechanical modeling, core analysis, and preliminary fracture simulations—has been completed. However, several gaps remain that limit the confidence level for scaling up to full-field development.

Key deficiencies were identified in geological modeling, fracture validation, and fluid–rock interaction testing. The static reservoir model is currently based on a simplified grid with limited structural and stratigraphic heterogeneity, restricting volumetric accuracy and the ability to analyze inter-well connectivity. Similarly, the mechanical earth model (MEM) has primarily focused on limestone intervals, with insufficient testing of shale layers and no regional propagation with quantified uncertainty—both of which are critical for accurate stress profiling and fracture growth prediction.

Fracture design simulations have been run under multiple propagation scenarios, but field validation of propped fracture geometry and stage contribution remains pending. While temperature logs confirm overall fracture containment, they lack the resolution to differentiate individual stage performance. More advanced surveillance techniques—such as chemical tracers, radioactive tracers, production logging tools (PLT), and distributed fiber-optic monitoring (DAS/DTS)—are required to quantify stage-by-stage contribution, evaluate cluster efficiency, and refine fracture placement strategies. Without these diagnostics, the link between modeled geometry and actual inflow distribution remains uncertain.

Laboratory testing also remains incomplete. While HPHT rheology and limestone mechanical testing have been performed, essential evaluations—such as retained conductivity, embedment under closure stress, proppant diagenesis, and shale–brine compatibility—are outstanding. These are particularly important given the mixed lithology of the reservoir and the potential for long-term fracture degradation. [Table 5](#) summarizes the current inputs and outstanding technical requirements. Addressing these gaps—through expanded geological modeling, multi-lithology MEM calibration, advanced laboratory testing, and high-resolution surveillance—will be essential to improve production forecasting, reduce uncertainty, and optimize future MSF well designs.

Specific to surveillance, limitations in post-DataFRAC temperature logs highlight the need for more accurate post-fracturing height measurements (e.g., sonic anisotropy analysis, radioactive tracers, non-radioactive tracers). Such techniques help determine how far the fracture has extended vertically across the layered formation and whether it has remained contained within the target zone. Understanding the vertical and lateral extent of fractures in laminated sequences is critical, as bedding planes can strongly influence fracture propagation behavior ([Cheng et al., 2022](#)).

Table 5—Current status and outstanding technical needs for SXX-ML development.

Domain	Current Inputs	Outstanding Needs
Geological Modeling	Simplified static grid; limited lateral heterogeneity; Upper 1–5 stratigraphy defined	Full 3D static model with structural detail; improved volumetrics; inter-well connectivity analysis
Geomechanics (MEM)	1D MEM calibrated to Pilot logs and limestone core; stress orientation defined	Multi-lithology MEM (limestone + shale); regional propagation; uncertainty quantification
Fracture Validation	Fracture simulations completed; temp logs confirm overall containment	Propped fracture diagnostics (PBU, PLT, chemical and radioactive tracers, fiber optics); cluster efficiency
Surveillance	Post-injection temperature logs in pilot well	Bottomhole pressure (BHP) analysis in pilot wells; extended well testing (EWT) for dynamic reservoir validation; accurate fracture geometry measurement using tracers or sonic anisotropy; alignment of decline analysis, logs, PBU, and production data; distributed fiber-optic monitoring (DAS/DTS); production logging (PLT); and tracer-based surveillance for stage contribution.
Laboratory Testing	HPHT rheology; compatibility testing	Retained conductivity; proppant embedment.
Reservoir Simulation	Preliminary models with pseudo-3D MEM	History-matched full-field simulation incorporating fracture complexity and depletion effects

Summary

This study presents a cost-effective workflow to optimize multistage hydraulic fracturing (MSF) in Pakistan's heterogeneous carbonate gas reservoirs, where full 3D Mechanical Earth Model (MEM) data are lacking. By integrating a core-calibrated 1D MEM with a 3D static grid and anchoring the design to pilot diagnostics, the approach reduced uncertainty in fracture prediction and production forecasting. The vertical pilot well confirmed fracture growth across the most productive Geological Flow Unit (GFU-3), while analogue benchmarking from the Pilot well reinforced the adoption of channel fracturing and conductivity retention strategies.

Key outcomes include:

- **Workflow Integration:** 1D MEM calibrated to core and fracture diagnostics was propagated into a pseudo-3D model, enabling realistic fracture geometry prediction within the pilot offset zone.
- **MSF Design:** A slanted lateral was planned with stage and cluster placement guided by Reservoir and Completion Quality Indices (RQI/CQI), ensuring stimulation focused on productive GFU-3 intervals. Channel fracturing with dual proppant (spherical ISP + rod-shaped) was adopted to enhance conductivity and mitigate flowback.
- **Production Forecasting:** Sector dynamic simulation projected initial post-frac rates and cumulative recovery over three years, depending on drawdown management.
- **Insights:** Vertical pilot wells and analogue benchmarking proved critical for model calibration; pressure sensitivity and convergence effects emerged as decisive factors in long-term performance.

While results demonstrate strong potential for scaling MSF development, the study also highlights remaining gaps—particularly in high-resolution fracture surveillance, multi-lithology MEM calibration, and advanced laboratory testing. Addressing these will be essential for reliable full-field deployment.

Acknowledgments

The authors thank Oil and Gas Development Company Limited (OGDCL) and SLB Reservoir Performance Services Pakistan for permission to publish this work and for their continued support in advancing hydraulic fracturing solutions in Pakistan. Special appreciation is extended to the field operations and subsurface teams for their efforts in data acquisition, pilot execution, and post-treatment surveillance. The authors also acknowledge the contributions of technical reviewers and collaborators whose insights strengthened the quality of this study.

Reference

- Abbas, H., Hedayati, S., and Meadows, D.L. 1992. Nonplanar Fracture Propagation from a Horizontal Wellbore: Experimental Study. *SPE Production & Facilities* **11** (3): 133–137. Paper SPE-24823-PA, presented at the SPE Annual Technical Conference and Exhibition, Washington, D.C., USA, 4–6 October. <https://doi.org/10.2118/24823-PA>
- Ali, M., Saqib, C.M., Rizvi, S.M.I., Hamim, N., Javed, A., and Shah, S.U. 2023. The Characterization of Geological Flow Units in Carbonate Reservoirs – Integration of Core, Log & Production Logging Data. Paper SPE-217362-MS presented at the SPE Middle East Oil and Gas Show and Conference, Manama, Bahrain, 19–21 February.
- Al Maki, A., Kurniadi, S.D., Gomaa, A.M., Lutfi, S., Ahmed, H., Al Darai, A., Al Fadhli, K., Casero, A., Rylance, A., and Al Shuraiqi, K. 2025. Pillar Frac Strategy in BP Oman: Case Studies of Implementation in Barik and Forecasted Challenges in Miqrat. Paper SPE-224884-MS presented at the SPE Conference at the Oman Petroleum & Energy Show, Muscat, Oman, 12–14 May. <https://doi.org/10.2118/224884-MS>
- Babatunde, A., Ejofodomi, E., and Adeoti, O. 2013. Optimizing Perforation Cluster Efficiency in Horizontal Wells Using Reservoir and Completion Quality Indices. Paper SPE-167128-MS presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, 30 September–2 October. <https://doi.org/10.2118/167128-MS>
- Ejofodomi, E., Babatunde, A., and Adeoti, O. 2014. Improving Cluster Contribution in Tight Reservoirs Through Integrated Petrophysical and Geomechanical Screening. Paper SPE-170996-MS presented at the SPE Annual

- Technical Conference and Exhibition, Amsterdam, The Netherlands, 27–29 October. <https://doi.org/10.2118/170996-MS>
- Chang, C., Zoback, M.D., and Khaksar, A. 2006. Empirical Relations Between Rock Strength and Physical Properties in Sedimentary Rocks. *J. Pet. Sci. Eng.* **51** (3–4): 223–237.
- Cheng, S., Sheng, M., and Deng, C. 2022. Identification of Different Lithofacies Laminations in Oil Shale and Their Mechanical Properties. *National Key Laboratory of Petroleum Resources and Engineering*, China University of Petroleum (Beijing), Beijing, China.
- Cipolla, C.L., Lonon, E., Mayerhofer, M.J., and Porterfield, K. 2011. Completion Advisor: A Workflow for Fracture Design Optimization. Paper SPE-140428-MS presented at the SPE Hydraulic Fracturing Technology Conference, The Woodlands, Texas, USA, 24–26 January. <https://doi.org/10.2118/140428-MS>
- El Rabaa, W. 1989. Experimental Study of Hydraulic Fracture Geometry Initiated from Horizontal Wells. Paper SPE-19720-MS presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, 8–11 October. <https://doi.org/10.2118/19720-MS>
- Horsrud, P. 2001. Estimating Mechanical Properties of Shale From Empirical Correlations. *SPE Drill. & Compl.* **16** (2): 68–73. SPE-56017-PA. <https://doi.org/10.2118/56017-PA>
- Lacy, L.L. 1997. Dynamic-to-Static Modulus Correlation for Shales and Weak Rocks. Presented at the 38th U.S. Rock Mechanics Symposium, Washington, D.C., USA, 7–10 July. ARMA-97-0437.
- McDaniel, G., Abbott, J., Mueller, F., Mokhtar, A., Pavlova, S., Neuvonen, O., Parias, T., and Alary, J.A. 2010. Changing the Shape of Fracturing: New Proppant Improves Fracture Conductivity. Paper SPE-135360-MS presented at the SPE Annual Technical Conference and Exhibition, Florence, Italy, 19–22 September. <https://doi.org/10.2118/135360-MS>
- Nicolaysen, A., Casero, A., Roy, A., Rylance, M., Kurniadi, S., and Batmaz, T. 2015. The Good, the Bad and the Ugly: A Case History of a Multi-Stage Hydraulic Fracturing Horizontal Well Tight-Gas Development in Oman. Paper SPE-179142-MS presented at the SPE/IADC Middle East Drilling Technology Conference and Exhibition, Abu Dhabi, UAE, 26–28 January. <https://doi.org/10.2118/179142-MS>
- Siddiqui, H.A., Iqbal, M.S., Rafique, H., Hasnain, A., Ashraf, Z., Hamim, N., Hussain, S.K., and Haider, S.A. 2023. Successful Integration of Channel Fracturing Unveils Promising Prospects in a Southern Region of Pakistan. Paper SPE-219502-MS presented at the SPE Middle East Oil and Gas Show and Conference, Manama, Bahrain, 19–21 February.
- Walker, R., Cipolla, C., and Mack, M. 2012. Optimizing Cluster Spacing and Fracture Design in Horizontal Wells. Paper SPE-159154-MS presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, 8–10 October. <https://doi.org/10.2118/159154-MS>